

# Mathematics A

## Unit 2 • Chapter OL-T39 Classified

Cross-session • chapter-organised candidate

2 classified items • 8 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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# Contents

Unit 2 • Unit 2 • Chapter OL-T39 • 2 items • 8 marks

| Chapter                              | Items | Marks | Starts |
|--------------------------------------|-------|-------|--------|
| 01. OL-T39 Standard & Compound Units | 2     | 8     | 4      |

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

# Standard & Compound Units

Unit 2 • 8 marks • 2 item(s)

January 2020 | 4MA1-2H Q2 | 2 marks

Convert cubic metric units and capacity

November 2025 | 2H Q26 | 6 marks

Make a sensible estimate of a measure

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

2 Change  $32.4 \text{ m}^3$  into  $\text{cm}^3$

.....  $\text{cm}^3$

## Worked solution

January 2020 | Question 2 | 2 marks

Family: Select and convert standard metric measures • Variant: Convert cubic metric units and capacity

Method / working

1.  $1 \text{ m}^3 = 100^3 = 1,000,000 \text{ cm}^3$ .

2.  $32.4 \times 1,000,000 = 32,400,000 \text{ cm}^3$ .

Final answer:  $32,400,000 \text{ cm}^3$

26 A solid cone is joined to a solid hemisphere to make the solid shown below.

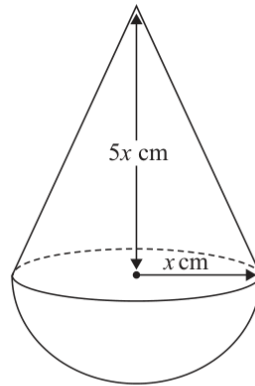


Diagram **NOT**  
accurately drawn

The cone is made from copper.  
The density of copper is  $9 \text{ g/cm}^3$

The hemisphere is made from a different metal.

The total mass of the solid is  $4752\pi$  grams  
The total volume of the solid is  $504\pi \text{ cm}^3$

Work out the density of the hemisphere.  
Show your working clearly.



..... g/cm<sup>3</sup>

(Total for Question 26 is 6 marks)

## Worked solution

November 2025 | Question 26 | 6 marks

Family: Read time and measuring instruments • Variant: Make a sensible estimate of a measure

Method

Volume of cone:

$$\frac{1}{3}\pi x^2(5x) = \frac{5}{3}\pi x^3$$

Volume of hemisphere:

$$\frac{2}{3}\pi x^3$$

Total volume:

$$\frac{5}{3}\pi x^3 + \frac{2}{3}\pi x^3 = 504\pi$$

$$\frac{7}{3}\pi x^3 = 504\pi$$

$$x^3 = 216$$

$$x = 6$$

Cone volume:

$$\frac{5}{3}\pi(6)^3 = 360\pi$$

Mass of copper cone:

$$360\pi \times 9 = 3240\pi$$

Mass of hemisphere:

$$4752\pi - 3240\pi = 1512\pi$$

Hemisphere volume:

$$\frac{2}{3}\pi(6)^3 = 144\pi$$

Density of the hemisphere:

$$(1512\pi)/(144\pi) = 10.5$$

Answer: 10.5 g/cm<sup>3</sup>